

Classify an Unlabeled Ground Cover Image

[My Solutions >](#)

To classify a new, unlabeled image, you must extract predictor features from the image. The trained models can then assess these predictor features and make a class for the image.

If your function does not pass the assessment, go back to the project introduction reading and refine your approach in MATLAB.

You will be assessed on:

1. Predictor feature values for `gcTableSaturation.avgSat` and `gcTableSaturation.stdSat`
2. Predicted classification of the unlabeled image in `gcTableSaturation.prediction`
3. Predictor feature values for `gcTableBag.f1` through `gcTableBag.f500`
4. Predicted classification of the unlabeled image in `gcTableBag.prediction`

We are providing you with the following:

- `img` - the unlabeled image in the RGB color space.
- `gcClassifierSaturation` - an SVM model trained to classify ground cover images using hand-selected predictor features based on image saturation values
- `gcClassifierBag` - an SVM model trained to classify ground cover images using predictor features automatically generated using the `bagOfFeatures` function
- `bag` - the bag of visual words object used to encode the predictor features used to train `gcClassifierBag`.

Script ?

 Save

 Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 img = imread("ocean.jpg"); % Load the unlabeled image
2 load gcClassifierSaturation.mat % Load the model trained using hand-selected features
3 load gcClassifierBag.mat % Load the model trained using automatically generated features
4 load bag.mat % Load bag of visual words object created by bagOfFeatures
5
6 % Paste your code from "predictUnlabeledGroundCoverImage.mlx" here:
7 % Step 1: Create a table of saturation-based predictor features
8 img = im2double(img);
9 imgHSV = rgb2hsv(img);
10 imgSaturation = imgHSV(:,:,2);
11 avgSat = mean(imgSaturation(:));
12 stdSat = std(imgSaturation(:));
13 gcTableSaturation = table(avgSat,stdSat);
14
15 % Step 2: Create a table of bagOfFeatures-based predictor features
16 predFeats = encode(bag, img);
17 featNames = "f" + string(1:500);
18 gcTableBag = array2table(predFeats,"VariableNames",featNames);
19
20 % Step 3: Use gcClassifierSaturation.predictFcn to classify the unlabeled
21 % image
22 gcTableSaturation.prediction = gcClassifierSaturation.predictFcn(gcTableSaturation);
23
24 % Step 4: Use gcClassifierBag.predictFcn to classify the unlabeled image
25 gcTableBag.prediction = gcClassifierBag.predictFcn(gcTableBag);
```

 Run Script

?

Assessment: All Tests Passed

Submit

?

✓ Saturation-Based Hand-Selected Features are Extracted Correctly

✓ Features Generated Using Bag of Features are Correct

✓ Check Model Predictions

Output

```
Encoding images using Bag-Of-Features.
-----
* Encoding an image...done.
```

